

Muhammad SHAHEER

Research Scientist | Digital Twins • Knowledge Graphs • Computer Vision • Robotics

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PROFESSIONAL SUMMARY

Research scientist (Ph.D. Computer Science – Robotics; M.Sc. Robot Systems) with 5 years of experience designing and developing **real-time digital twin applications** of the built environment. Co-developer of **S-Graphs**, an open-source framework that turns live LiDAR, camera and IMU data into a **hierarchical semantic knowledge graph** of buildings, and of methods integrating **BIM/IFC models and architectural plans** with sensor data to keep the virtual model aligned with the as-built reality. Experienced in combining **first-principle, probabilistic/statistical, data-driven (GNN) and computer vision models**, in **uncertainty quantification for risk-aware decision support**, and in shipping reproducible software (C++, Python, Git, Docker). 10 peer-reviewed publications (8 IEEE RA-L, 2 IROS), 2nd place in the Hilti x Trimble 2026 Localization Challenge, and a track record of international conference presentations and cross-functional collaboration.

PROFESSIONAL EXPERIENCE

- **Post-Doctoral Researcher – Digital Twins & Spatial AI** Oct 2025 – Present
University of Luxembourg, SnT – Automation & Robotics Research Group Kirchberg, Luxembourg
 - Lead research on **digital-twin-ready 3D scene graphs** that link real-time robot sensor data with **BIM/IFC** building information models, keeping the virtual representation synchronized with the as-built state of construction sites and indoor facilities.
 - Design **semantic frameworks** (asset hierarchies building → floor → room → wall/structural element, graph-based ontologies and knowledge graphs) enabling **interoperability and advanced inference across heterogeneous data sources** (LiDAR, RGB-D, IMU, BIM, floor plans).
 - Integrate **probabilistic and statistical models** (factor graphs, uncertainty-aware landmarks) with **data-driven models** (Graph Neural Networks) and **computer vision**, quantifying confidence to detect deviations between planned and as-built models and support **risk evaluation and decision making**.
 - Benchmarked the pipeline against industrial ground truth in the **Hilti x Trimble SLAM Challenge 2026**, placing **2nd in Localization** and 5th in SLAM.
- **Doctoral Researcher – Robotics & Digital Representations of Buildings** Nov 2021 – Sep 2025
University of Luxembourg, SnT – Automation & Robotics Research Group Kirchberg, Luxembourg
 - Co-developed **S-Graphs, S-Graphs+ and S-Graphs 2.0**, an open-source, **real-time** framework that builds a multi-layer **hierarchical knowledge graph** (sensor, geometric, room and floor layers) from streaming LiDAR data, with hierarchical optimization and loop closure for large, multi-floor buildings (IEEE RA-L 2022, 2023, 2025).
 - Developed **graph-matching methods** that align robot-built situational graphs with **architectural graphs extracted from BIM and floor plans**, enabling global localization and tightly coupled SLAM even when plans are imprecise (IROS 2023, Robotics 2023, IEEE RA-L 2025).
 - Contributed to **Graph Neural Network** models that infer high-level semantic concepts (rooms, walls) with **uncertainty estimates** from low-level geometry, combining data-driven and first-principle geometric models (IROS 2024, IEEE RA-L 2026).
 - Contributed to **BIM-informed Visual SLAM** for construction environments and **situationally-aware path planning** that exploits the semantic graph for safer, context-aware decisions (IEEE RA-L 2026).
 - Engineered research software in **C++ and Python** on **ROS/ROS 2**, with **Git** version control and **Docker** containerization for reproducible deployment; validated in simulation (Gazebo, Unity) and on real-world industrial/construction datasets.
 - Presented results at IEEE/RSJ IROS and IEEE ICRA workshops (3 awards) and collaborated with international partners (University of Zaragoza, University of Amsterdam / ETH Zurich).

TECHNICAL SKILLS

- **Digital Twin & Semantic Modeling:** Digital twins of buildings and industrial sites, 3D scene graphs, knowledge graphs, ontologies, asset hierarchies, BIM / IFC, semantic interoperability, heterogeneous data integration
- **Modeling Approaches:** First-principle geometric/kinematic models, probabilistic & statistical models (factor graphs, Kalman and particle filters, uncertainty quantification), data models (Graph Neural Networks, CNNs), computer vision
- **Real-time Data & Monitoring:** Real-time sensor fusion of streaming LiDAR (Velodyne, Ouster), IMU and RGB-D (Intel RealSense) data, SLAM, localization, deviation detection, pose graph optimization
- **Programming & Frameworks:** Python, C++, ROS / ROS 2, OpenCV, PCL (Point Cloud Library), Eigen, g2o; simulation and 3D visualization with Gazebo, Unity, RViz

- **Software Engineering & DevOps:** Git version control, Docker, Linux, reproducible deployment, open-source development
- **Professional:** Analytical problem solving, cross-functional collaboration, scientific writing, conference presentations, student supervision

EDUCATION

- **Ph.D. Computer Science (Robotics)**
University of Luxembourg – Thesis on semantic 3D scene graphs informed by architectural/BIM models Kirchberg, Luxembourg
- **M.Sc. Robot Systems**
University of Southern Denmark – Danish Government Scholarship Odense, Denmark
- **B.E. Mechatronics Engineering**
National University of Sciences and Technology (NUST) Islamabad, Pakistan

HONORS, AWARDS & MEMBERSHIPS

- **2nd Place, Localization Challenge (5th, SLAM Challenge)** – Hilti x Trimble SLAM Challenge 2026 Jun 2026 [\[link\]](#)
- **Best Presentation Award Finalist** – ICRA 2026 Robot Meets Prior Maps Workshop Jun 2026 [\[link\]](#)
- **Best Poster Award, 2nd Place** – ICRA 2025 Future of Construction Workshop 2025 [\[link\]](#)
- **Best Research Award, 2nd Place** – ICRA 2023 Future of Construction Workshop Jun 2023 [\[link\]](#)
- **Danish Government Scholarship** – full tuition waiver and stipend, M.Sc. Robot Systems 2019 – 2021 [\[link\]](#)
- **Professional Memberships** – IEEE Member; IEEE Robotics & Automation Society; IEEE Young Professionals

KNOWLEDGE TRANSFER, TEACHING & SUPERVISION

- **Summer School Lecture** – hierarchical LiDAR SLAM, hands-on Docker tutorial Jul 2026 [\[link\]](#)
- **M.Sc. Thesis Supervision** – open-source multimodal sensing platform Sep 2024 – Jun 2025 [\[link\]](#)
- **B.Sc. Projects Supervision** – BIM data extraction; *as-built* IFC from robot data Sep 2023 – Jun 2024

PUBLICATIONS

J=JOURNAL, C=CONFERENCE, S=IN SUBMISSION

- [J.1] A. Bikandi, M. Cortizas, M. Shaheer et al. (2026). **BIM Informed Visual SLAM for Construction Environments**. *IEEE Robotics & Automation Letters (RA-L)*, Early Access.
- [J.2] J.A. Millan-Romera, M. Shaheer et al. (2026). **Generation of Uncertainty-Aware High-Level Spatial Concepts in Factorized 3D Scene Graphs via Graph Neural Networks**. *IEEE RA-L*, 11(10), 11243–11250.
- [J.3] S. Ejaz, M. Giberna, M. Shaheer et al. (2026). **Situationally-aware Path Planning Exploiting 3D Scene Graphs**. *IEEE RA-L*, 11(3), 3358–3365.
- [J.4] H. Bavle, J.L. Sanchez-Lopez, M. Shaheer et al. (2025). **S-Graphs 2.0 – A Hierarchical-Semantic Optimization and Loop Closure for SLAM**. *IEEE RA-L*, 10(12), 12461–12468.
- [J.5] M. Shaheer et al. (2025). **Tightly Coupled SLAM With Imprecise Architectural Plans**. *IEEE RA-L*, 10(8), 8019–8026.
- [J.6] M. Shaheer et al. (2023). **Robot Localization Using Situational Graphs (S-Graphs) and Building Architectural Plans**. *Robotics (MDPI)*, 12(3), 65.
- [J.7] H. Bavle, J.L. Sanchez-Lopez, M. Shaheer et al. (2023). **S-Graphs+: Real-Time Localization and Mapping Leveraging Hierarchical Representations**. *IEEE RA-L*, 8(8), 4927–4934.
- [J.8] H. Bavle, J.L. Sanchez-Lopez, M. Shaheer et al. (2022). **Situational Graphs for Robot Navigation in Structured Indoor Environments**. *IEEE RA-L*, 7(4), 9107–9114.
- [C.1] J.A. Millan-Romera, M. Shaheer et al. (2024). **Learning High-level Semantic-Relational Concepts for SLAM**. *IEEE/RSJ IROS 2024*, Abu Dhabi, UAE.
- [C.2] M. Shaheer et al. (2023). **Graph-Based Global Robot Localization Informing Situational Graphs with Architectural Graphs**. *IEEE/RSJ IROS 2023*, Detroit, USA.
- [S.1] M. Giberna, M. Shaheer et al. (2026). **Robust SLAM with 3D Scene Graph Using Uncertain Dynamic Landmarks**. Submitted to *IEEE RA-L*.

REFERENCES

- **Prof. Dr. Holger Voos** (Ph.D. Advisor)
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